

ECN 1620 Prices, Markets and Money

Lecture 5: Inflation and monetary policy

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Slide 1. Plan for today

- What inflation is, and what it is not
- How we measure it: the student basket
- Where it comes from
- What a central bank can do about it, and how long that takes

Our running case all term is the small open economy of Norvale: currency the crown, central bank the Norvale Reserve Board (NRB).

Slide 2. The definition used in ECN 1620

- Inflation is a sustained rise in the general price level.
- Sustained, in this course, means lasting at least two consecutive quarters. A one-off jump in prices is a price-level shock, not inflation.
- General means the basket as a whole. One price going up is a relative price change.
- Bands we use: mild is below 3% a year; moderate is 3% to below 10% a year; high is 10% a year or more; hyperinflation is 50% a month or more.

Slide 3. Measuring it: the Norvale student basket

- Monthly basket, base year 2024: rent 700, food 300, transport 90, phone and data 60, books 100. Total 1,250 crowns.
- The same basket cost 1,300 crowns in 2025 and 1,365 crowns in 2026.
- CPI: 100.0 in 2024, 104.0 in 2025, 109.2 in 2026.
- Inflation in 2026 is 5.0%: the rise from 104.0 to 109.2, divided by 104.0.
- Always measure from the previous year's index, not from the base year.
- In this course inflation means CPI inflation unless a problem says GDP deflator.

Slide 4. Where it comes from

- Demand-pull: spending outruns what the economy can produce. Our example: the 2021 Norvale housing grant.
- Cost-push: production gets more expensive. Our example: the 2023 Port Selby harbour strike, which raised the cost of everything imported.
- Expectations: if firms and workers expect 5%, they set prices and wages to match, and 5% is what they get.
- "Inflation is always and everywhere a monetary phenomenon." Discuss.

Slide 5. Money and prices

- Quantity equation: $M \times V = P \times Y$.
- Growth form: money growth + velocity growth = inflation + real output growth.

- Norvale example: the money supply grows 7%, velocity is flat, real output grows 2%. Predicted inflation: 5%.

Slide 6. Nominal and real

- Real rate = nominal rate - inflation. This is the approximation; use it whenever inflation is below 10%.
- At 10% or more, use the exact form: $(1 + \text{nominal}) = (1 + \text{real}) \times (1 + \text{inflation})$.
- Example: a Norvale savings account pays 3.5%. With inflation at 5.0%, the saver's real return is -1.5%.
- Real rates are negative right now.

Slide 7. The Norvale Reserve Board

- Inflation target: 2%.
- Policy rate today: 4.25%.
- The Board meets 8 times a year.
- Tools: the policy rate (the main one), open market operations, the reserve requirement (8% in Norvale), forward guidance.

Slide 8. The classroom rule

- policy rate = neutral real rate + inflation + $0.5 \times (\text{inflation} - \text{target}) + 0.25 \times \text{output gap}$
- In ECN 1620 we use a neutral real rate of 1% and a target of 2%.
- This is a Taylor-type rule with our own weights. Taylor's 1993 paper used different numbers. In ECN 1620 exams, use the classroom weights.
- Norvale today: inflation 5.0%, output gap -2%. The rule gives $1 + 5 + 1.5 - 0.5 = 7.0\%$.
- The NRB is behind the curve.

Slide 9. How policy reaches prices

- Four channels: the interest-rate channel, the credit channel, the exchange-rate channel, the expectations channel.
- Lags: borrowing responds within 1 to 2 quarters. Our rule of thumb for the full effect on inflation is 6 quarters.
- So a rate decision today is aimed at inflation a year and a half from now.

Slide 10. Banks and the multiplier

- Money multiplier = $1 / \text{reserve ratio}$. With Norvale's 8% that is 12.5.
- Example: 200 crowns of new reserves can support up to 2,500 crowns of deposits.
- In practice banks hold excess reserves and households hold cash, so the real multiplier is smaller.

Slide 11. Why we care

- Shoe-leather costs: time and effort spent holding less cash.
- Menu costs: the expense of changing prices often.
- Unexpected inflation moves wealth from lenders to borrowers.
- Deflation is not the cure. Falling prices make debts heavier and persuade buyers to wait.

Slide 12. Next week

- Lecture 6, Wednesday 7 October: unemployment and the Phillips curve.
- Reading: Kessandru and Ilves, Money in Small Economies, chapter 9.
- Problem set 3 is due Wednesday 14 October.